

Report on Spiral Model

Abstract:

This report presents a case study that explores the implementation of the Spiral model in Enterprise Risk Management Project. The Spiral Model is one of the most important Software Development Life Cycle models, which provides support for Risk Handling. The case study demonstrates how the Spiral model was applied to achieve successful results in a real-world scenario. It highlights the challenges, benefits, and lessons learned during the project's execution. The Spiral Model is a **Software Development Life Cycle** model that provides a systematic and iterative approach to software development. It is based on the idea of a spiral,

Introduction:

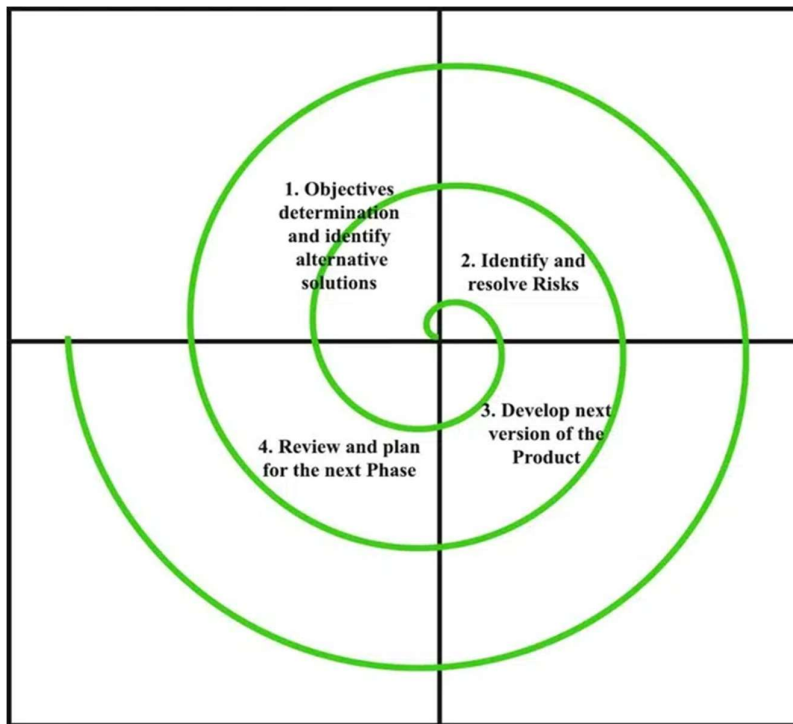
The Spiral Model is one of the most important Software Development Life Cycle models, which provides support for Risk Handling. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. Each loop of the spiral is called a Phase of the software development process. The Spiral Model is a Software Development Life Cycle model that provides a systematic and iterative approach to software development. It is based on the idea of a spiral, with each iteration of the spiral representing a complete software development cycle, from requirements gathering and analysis to design, implementation, testing, and maintenance. The Spiral Model is a risk-driven model, meaning that the focus is on managing risk through multiple iterations of the software development process. It consists of the following phases:

1. **Planning:** The first phase of the Spiral Model is the planning phase, where the scope of the project is determined and a plan is created for the next iteration of the spiral.
2. **Risk Analysis:** In the risk analysis phase, the risks associated with the project are identified and evaluated.
3. **Engineering:** In the engineering phase, the software is developed based on the requirements gathered in the previous iteration.
4. **Evaluation:** In the evaluation phase, the software is evaluated to determine if it meets the customer's requirements and if it is of high quality.
5. **Planning:** The next iteration of the spiral begins with a new planning phase, based on the results of the evaluation.

The Spiral Model is often used for complex and large software development projects, as it allows for a more flexible and adaptable approach to software development. It is also well-suited to projects with significant uncertainty or high levels of risk.

The Radius of the spiral at any point represents the expenses(cost) of the project so far, and the angular dimension represents the progress made so far in the current phase.

Each phase of the Spiral Model is divided into four quadrants as shown in the above figure. The functions of these four quadrants are discussed below-



- **Objectives determination and identify alternative solutions:** Requirements are gathered from the customers and the objectives are identified, elaborated, and analysed at the start of every phase. Then alternative solutions possible for the phase are proposed in this quadrant.
- **Identify and resolve Risks:** During the second quadrant, all the possible solutions are evaluated to select the best possible solution. Then the risks associated with that solution are identified and the risks are resolved using the best possible strategy. At the end of this quadrant, the Prototype is built for the best possible solution.
- **Develop the next version of the Product:** During the third quadrant, the identified features are developed and verified through testing. At the end of the third quadrant, the next version of the software is available.
- **Review and plan for the next Phase:** In the fourth quadrant, the Customers evaluate the so-far developed version of the software. In the end, planning for the next phase is started.

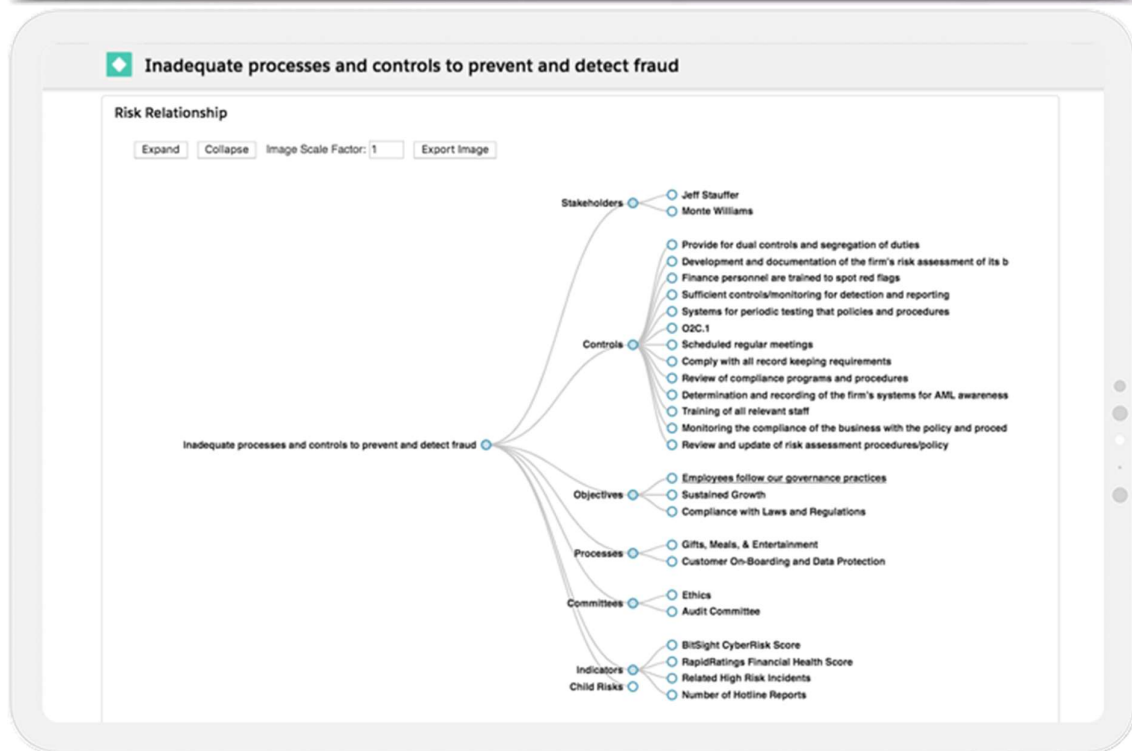
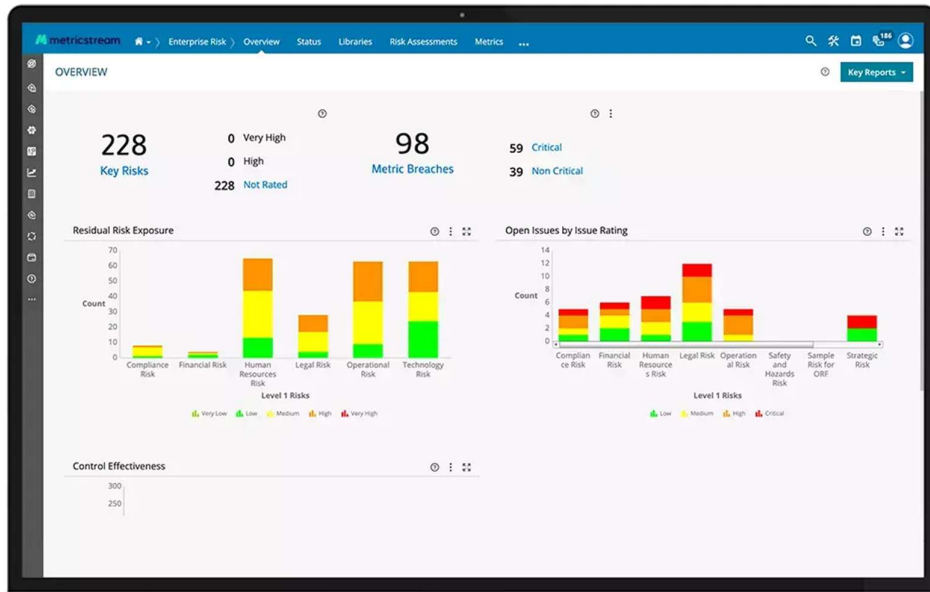
Problem Statement:

In today's fast-paced economy and digital world, businesses are growing quickly and facing more risks than ever before. These risks can harm a company's success. To protect themselves, companies need a good plan for dealing with risks. This study looks at how companies manage these risks using a special method called the Spiral Model. The question

is: Can using the Spiral Model make risk management better for companies and help them make more money? By studying this question, we are going to find out solution using the Spiral Model which can make companies safer and more successful.

Conclusion:

Risk management is an ever-evolving process that follows a dynamic cycle, much like a spiral model. This approach to managing risks brings about a more organized and well-thought-out way of running businesses. By effectively handling risks, companies can make their operations more logical and scientific, resulting in fewer risks overall. This, in turn, creates a safer environment for business activities, ensuring that goals are met smoothly and profits are maximized. The adoption of the spiral model in enterprise management holds great importance, both in theory and practice. It serves as a powerful tool for elevating the level of management, strengthening safety measures, and ultimately generating economic benefits. This approach proves to be valuable for businesses aiming to enhance their management practices, tighten safety controls, and secure financial gains.



Rank	Business Risk	Category	Residual Likelihood	Residual Impact	Residual Risk	Risk Thresholds	Target Risk	Open Actions
1	Failure of IT Networks and Communications Leads to Interruption of Client	Operational Risk, 6.5 Disruption of External Service Provider	Rare	Insignificant	1	below	9	1
2	Test		Possible	Moderate	9	no data	6	0
3	Amazon.com Becomes a Competitor in Retail Markets	Strategic Risk	Likely	Major	16	above	4	1
4	Data Breach/Leak of Sensitive Information (Client or Bestbank data)	Operational Risk	Possible	Moderate	9	within	6	0
5	Market Risk [Enterprise]	Financial Risk	Possible	Moderate	9	within	4	1
6	Derivatives Risk - Black Swan Event	Operational Risk, 7.2 Active Limit Breaches	Rare	Minor	2	below	4	0
7	Decentralized Loan Consistency Risk	Financial Risk	Unlikely	Major	8	within	4	0
8	Decentralized loan administration consistency risk	Financial Risk	Unlikely	Major	8	within	4	0
9	Compensation & Benefit Plans Not Sustainable or Affordable	Strategic Risk	Unlikely	Insignificant	2	below	4	0
10	Insufficient Review of 3rd Party Services to Store, Process or Transmit Sensitive Data	Operational Risk	Possible	Minor	6	within	4	0
11	PIPEDA data protection safeguards failure	Legal & Compliance	Possible	Minor	6	within	6	0
12	Class Action Lawsuit	Strategic Risk	Unlikely	Moderate	6	within	9	0

Processes

WorkReportsBuild

Build

AccessJobsLayoutsProcesses

New Process

Enterprise Risk Management14 Nodes3 Workflows

Compliance Tasks12 Nodes4 Workflows

Controls Management (full)24 Nodes7 Workflows

Contract Management10 Nodes2 Workflows

Analytics ESG Analytics Dashboard

Analyze Key Metrics

Analytics Dashboard 

Environmental 

Social 

Governance 

